A

BILEVEL OPTIMISATION

1,300+ optimisation runs
→ 93 unique scenarios

Bio_Igg: 53 | Igg_Igg: 40
Average: 130 regulations/scenario

Sensitivity Analysis

REGULATION REDUCTION

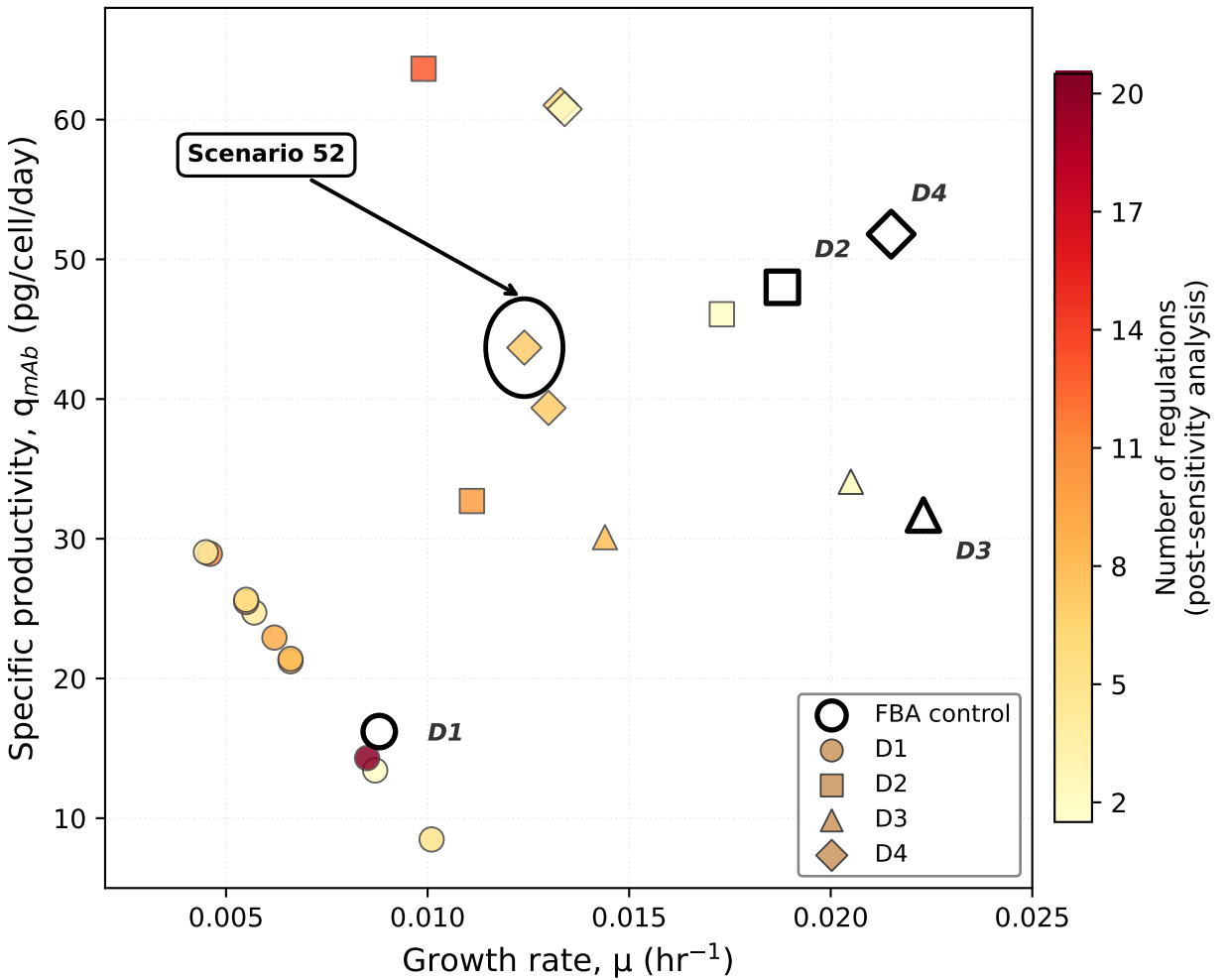
93 → 89 scenarios
(4 duplicates removed)

130 → 9 regulations (avg)
• All downregulations removed

μ >50% AND q_{mAb} >50%

HIGH-PRODUCING SCENARIOS

89 → 20 scenarios (22%)

Average: 5.6 regulations
Range: 2–20 regulations
→ Ready for analysis

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